Fantasy Football Recommender

V1 | REVISED BASELINE

Realistic V1 delivery plan | engineering timeline and release reserve

PROJECT TARGET
20 weeks

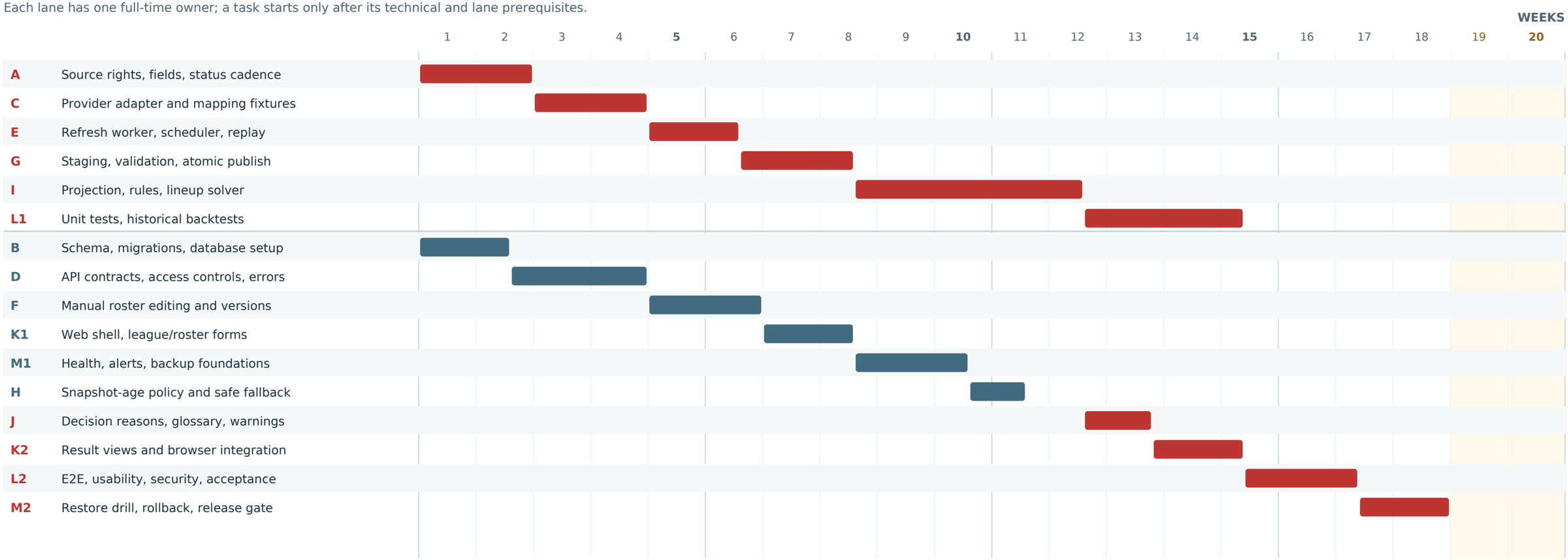
ENGINEERING
18 weeks / Day 90

CONTINGENCY
2 weeks / 10 days

CAPACITY ASSUMED
2 full-time equivalents

RESOURCE-LEVELED WORK PACKAGES

Each lane has one full-time owner; a task starts only after its technical and lane prerequisites.



RED: zero float

BLUE: capacity/precedence flexibility

AMBER: explicit release contingency (Weeks 19-20)

TWO EQUALLY LONG CRITICAL BRANCHES

A - C - E - G - I - J - K2 - L2 - M2

A - C - E - G - I - L1 - L2 - M2

Engineering Day 90 is the earliest complete V1 candidate; Day 100 is the planning target with a ten-day reserve.

Working days are relative to project start. Vendor approvals that exceed the reserve move the release date.

Dependency and resource audit

V1 | REVISED BASELINE

16 schedulable packages covering 13 core WBS components; Sleeper is outside the core schedule

ID	WBS	Work package	Owner	Technical preds	Lane prev.	Days	ES	EF	LS	LF	Float
A	1.3.2	Source rights, fields, status cadence	Data / ML	Start	-	10	0	10	0	10	0
B	1.3.1	Schema, migrations, database setup	App / Ops	Start	-	8	0	8	2	10	2
C	1.3.3	Provider adapter and mapping fixtures	Data / ML	A, B	A	10	10	20	10	20	0
D	1.2.1	API contracts, access controls, errors	App / Ops	B	B	12	8	20	13	25	5
E	1.2.4	Refresh worker, scheduler, replay	Data / ML	C	C	8	20	28	20	28	0
F	1.1.3	Manual roster editing and versions	App / Ops	D	D	10	20	30	25	35	5
G	1.2.5	Staging, validation, atomic publish	Data / ML	B, E	E	10	28	38	28	38	0
K1	1.1.1	Web shell, league/roster forms	App / Ops	D	F	8	30	38	35	43	5
I	1.2.2	Projection, rules, lineup solver	Data / ML	G	G	20	38	58	38	58	0
M1	1.4.2	Health, alerts, backup foundations	App / Ops	D, E, G	K1	10	38	48	43	53	5
H	1.2.3	Snapshot-age policy and safe fallback	App / Ops	G	M1	5	48	53	53	58	5
L1	1.4.1	Unit tests, historical backtests	Data / ML	G, I	I	14	58	72	58	72	0
J	1.1.2	Decision reasons, glossary, warnings	App / Ops	H, I	H	6	58	64	58	64	0
K2	1.1.1	Result views and browser integration	App / Ops	F, H, J, K1	J	8	64	72	64	72	0
L2	1.4.1	E2E, usability, security, acceptance	App / Ops	K2, L1	K2	10	72	82	72	82	0
M2	1.4.2	Restore drill, rollback, release gate	App / Ops	L2, M1	L2	8	82	90	82	90	0

WHY THIS NETWORK IS DIFFERENT

Resource constraints

The Data / ML and App / Ops lanes each allow one active full-time task. Lane predecessor links are capacity constraints; technical predecessors are engineering requirements.

Earlier integration

The web shell (K1) starts after API contracts. Results integration (K2) waits for the engine, explanations, freshness rules, and manual roster flow.

Two test and release gates

Historical backtests (L1) run alongside explanations and UI integration. End-to-end acceptance (L2) and recovery checks (M2) finish the baseline.

Critical path = 90 workdays; release reserve = 10 further workdays.

Float uses both technical and one-person-per-lane resource links. Dates do not imply a fixed calendar start.

Estimation basis and release gates

V1 | REVISED BASELINE

WBS-complete V1 scope, staffing assumptions, unresolved approvals, and measurable acceptance points

WORKLOAD AND CAPACITY

1.3.2	Permitted NFL feed	10
1.3.1	Database	8
1.3.3	Provider adapter	10
1.2.1	Application API	12
1.2.4	Scheduler/worker	8
1.1.3	Manual roster entry	10
1.2.5	Staging/validation	10
1.2.3	Freshness guard	5
1.2.2	Recommendation engine	20
1.1.2	Explanations/glossary	6
1.1.1	Web interface (K1 + K2)	16
1.4.1	Testing (L1 + L2)	24
1.4.2	Operations (M1 + M2)	18

ESTIMATE

157 person-days

1,256 planned labor hours

2 full-time-equivalent contributors

90 working days + 10 reserve days

52 required WBS outputs; 4 optional Sleeper outputs

At 1 FTE: at least 31.4 workweeks of labor

RELEASE CHECKPOINTS

W2	**Source approved**	Source-specific rights, sample coverage, and status cadence documented
W4	**Contracts established**	Core schema, provider mapping, and API boundary testable
W8	**Validated snapshot**	Staging rejects bad batches; atomic active-snapshot switch passes
W12	**Decision engine**	Deterministic scoring and eligibility rules work on real fixtures
W15	**Model and web integrated**	Historical backtests and results UI completed against the same decisions
W17	**Acceptance tests**	Manual roster-to-result E2E, access controls, failure flows, usability
W18	**Release candidate**	Backup restore, rollback, alerts, staleness protections, runbook pass
W20	**Delivery target**	Two-week schedule reserve for defects or short provider delays

ASSUMPTIONS AND RISKS

- Two full-time-equivalent lanes are available for 18 working weeks. At lower availability, relevel before committing to dates.
- Estimates are bottom-up engineering judgments, not observed team velocity. The 10-day reserve is excluded from CPM float.
- nflverse data may cover stats and daily injury updates, but validate the exact release license, fields, season, and status timing.
- Sleeper (1.3.4) is optional and unscheduled; commercial integration needs provider authorization. Waivers/trades/live updates are out of V1.
- External licensing or procurement waits longer than two weeks require a changed baseline. Hosting and alert providers remain TBD.

Project sources: Five-Level Technical WBS, Stack Overview presentation, original 21-day CPM. Provider checks: nflverse GitHub, nflreadr schedule, Sleeper API docs (Oct 2026).